

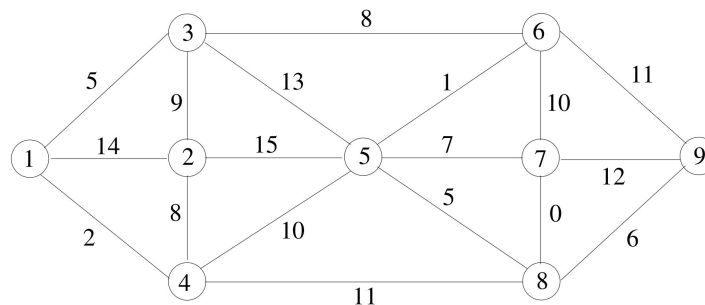
Take-home exam: December 19, 2012.

Course name: Network Optimization

Course no: 42115

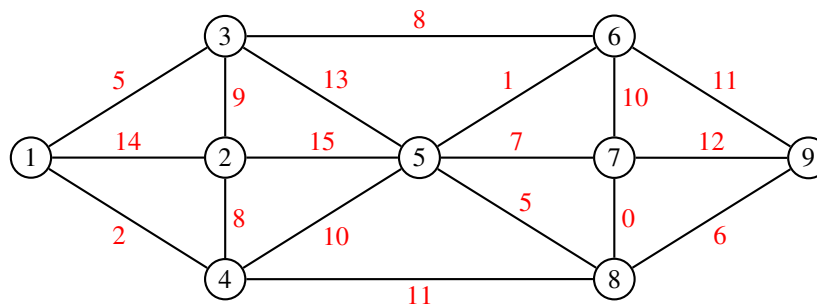
Minimum spanning tree

Given the following graph $G = (V, E)$



Problem 1: (small) Find the minimum spanning tree of G using Kruskal's algorithm ■

Answer 1



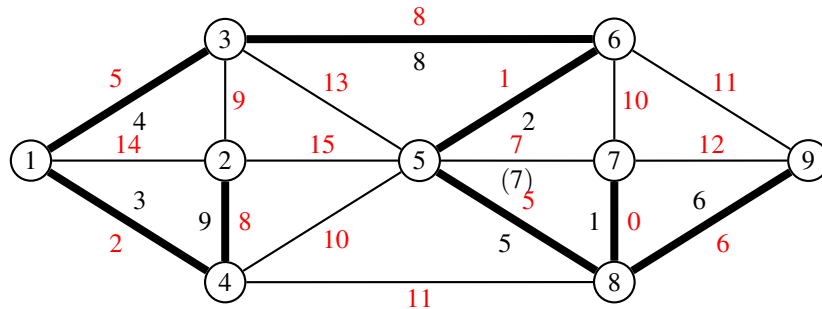
First we sort the edges according to their weight

$e_{78}, e_{56}, e_{14}, e_{13}, e_{58}, e_{89}, e_{57}, e_{36}, e_{24}, e_{23}, e_{45}, e_{67}, e_{48}, e_{69}, e_{79}, e_{35}, e_{12}, e_{25}$

We then add the edges in increasing order, each time checking if a cycle is introduced

$e_{78}, e_{56}, e_{14}, e_{13}, e_{58}, e_{89}, (e_{57}), e_{36}, e_{24}$

The edge marked with paranthesis was skipped since it would introduce a cycle. We stop when $n - 1 = 8$ edges has been added.



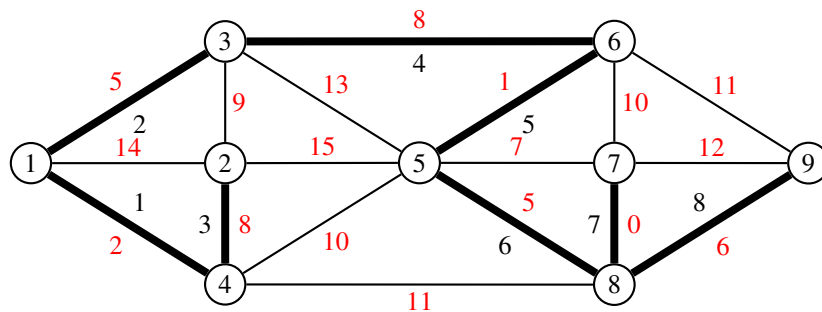
The cost of the MST is $0 + 1 + 2 + 5 + 5 + 6 + 8 + 8 = 35$. ■

Problem 2: (small) Find the minimum spanning tree of G using Prim's algorithm ■

Answer 2 Starting from node 1, we add the following edges:

$$e_{14}, e_{13}, e_{24}, e_{36}, e_{56}, e_{58}, e_{78}, e_{89}$$

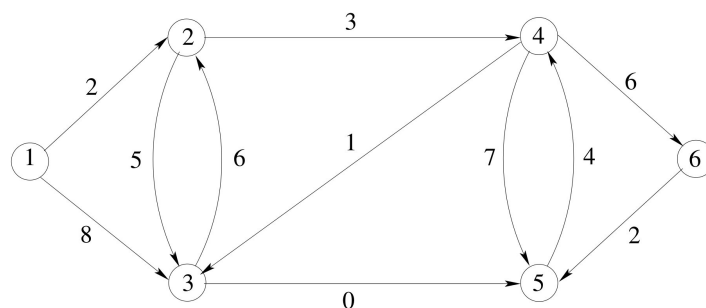
We stop when the last node has been considered.



The cost of the MST is again $0 + 1 + 2 + 5 + 5 + 6 + 8 + 8 = 35$. ■

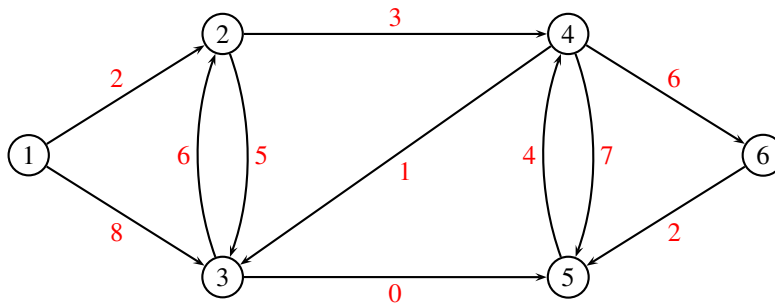
It is sufficient to indicate the chosen edges on the graph and to give each edge a sequence number (in which order they have been considered).

Shortest path

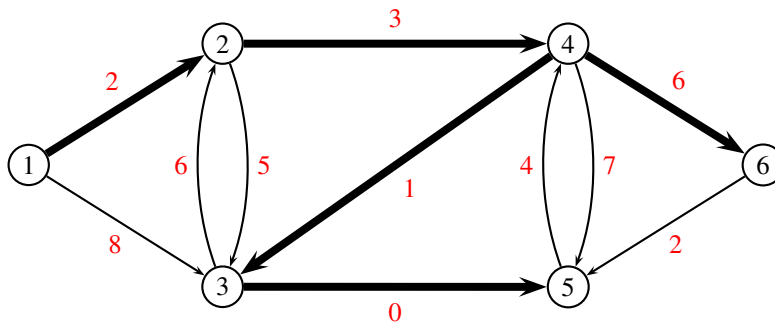


Problem 3: (small) Use Dijkstra's algorithm to solve the above shortest path problem starting from node 1. ■

Answer 3 We have the following



relaxing	potential					
	1	2	3	4	5	6
init	∞	∞	∞	∞	∞	∞
1	0	2	8	∞	∞	∞
2			7	5		
4			6		12	11
3					6	
5						
6						



It is sufficient to indicate the chosen edges in a shortest path tree on the graph and to give each edge a sequence number (in which order they have been considered). Moreover, the distance to each node should be indicated on the node.

Shortest path tree

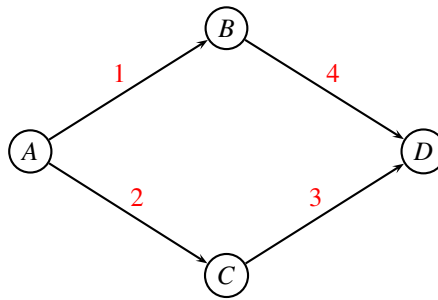
Problem 4: (large) Are the following statements correct? (Prove or disprove each statement. To disprove a statement it is sufficient to show an example.)

- The shortest path tree is unique when all edges have distinct weights.
- In a directed graph with positive edge weights the shortest paths do not change when directed edges are replaced with undirected edges.
- In a shortest path problem, if all edge weights are increased by adding a constant $k > 0$ then the cost of a shortest path is also increased by a multiple of k .

- d) Among all shortest paths in $G = (V, E)$ the Dijkstra algorithm finds the shortest path with fewest edges.

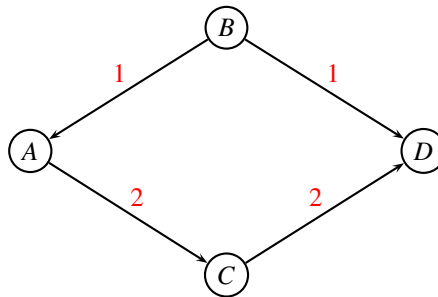
■

Answer 4 a) Consider the following weighted directed graph



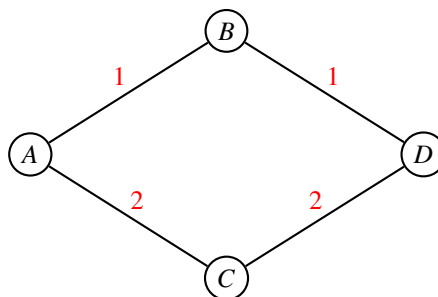
All edges have distinct weights, but there are two paths A to D having the same distance. So, the answer is: **false**.

b) Consider the following weighted directed graph



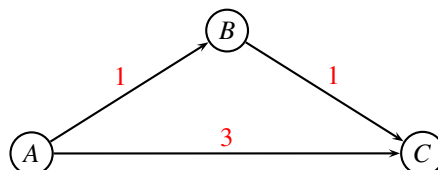
The shortest path from A to D is $A \rightarrow C \rightarrow D$.

Now, consider the same undirected graph



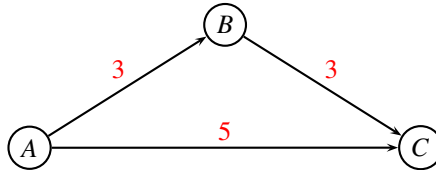
The shortest path from A to D is now $A \rightarrow B \rightarrow D$. So, the answer is: **false**.

c) Consider the following weighted directed graph



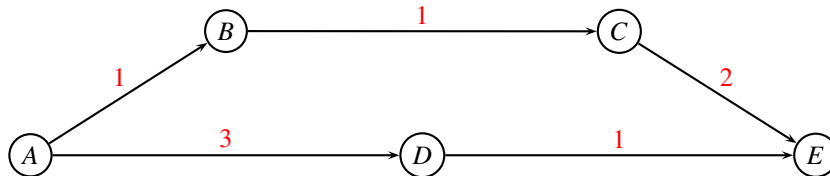
The shortest path from A to C is $A \rightarrow B \rightarrow C$ with cost 2.

Now, increase all weights by a constant $k = 2$



The shortest path from A to C is $A \rightarrow C$ with cost 5. The increase is not a multiple of $k = 2$. So, the answer is: **false**.

d) Consider the following weighted directed graph



Dijkstra will first consider node A , then node B and node C before considering node D . This means that it will find path $A \rightarrow B \rightarrow C \rightarrow E$ as shortest path from A to E . But the shortest path with minimum number of edges from A to E is $A \rightarrow D \rightarrow E$. So, the answer is: **false**.

e) Consider the same weighted directed graph as in d) above. Notice that Bellman-Ford does not define in which order the edges should be considered, we should just consider all edges. So, assume that Bellman-Ford in the first iteration considers the edges in the order

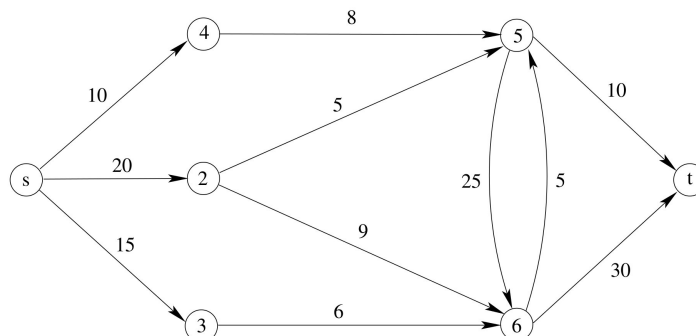
$(A, B), (B, C), (C, E), (A, D), (D, E)$.

This means that the path $A \rightarrow B \rightarrow C \rightarrow E$ is updated before path $A \rightarrow D \rightarrow E$, and hence we will not find the path with fewest edges. So, the answer is: **false**.

■

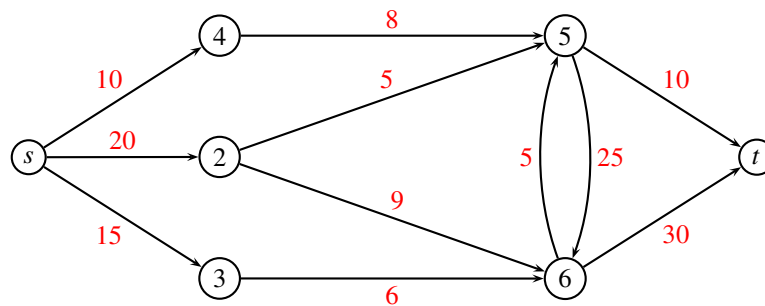
Maximum flow problem

Consider the maximum flow problem with source s and sink t



Problem 5: (medium) Solve the max flow problem using an augmenting path algorithm. In each iteration let the augmenting path be the shortest (measured in number of edges) path from s to t in the residual graph. In each iteration write down which augmenting path was used. ■

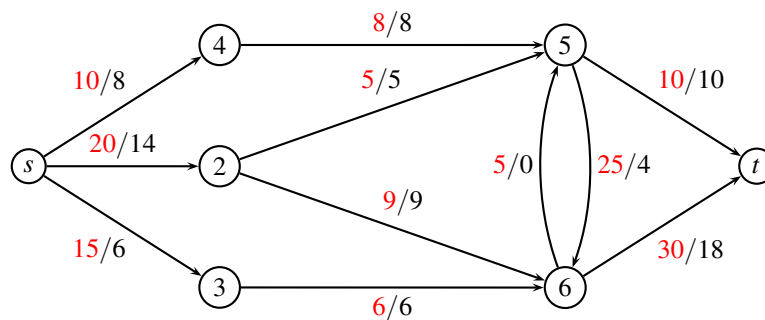
Answer 5 We have



We do the following iterations:

- Augmenting path: $s45t$, capacity 8.
- Augmenting path: $s26t$, capacity 9.
- Augmenting path: $s36t$, capacity 6.
- Augmenting path: $s25t$, capacity 2.
- Augmenting path: $s256t$, capacity 3.

Getting the solution



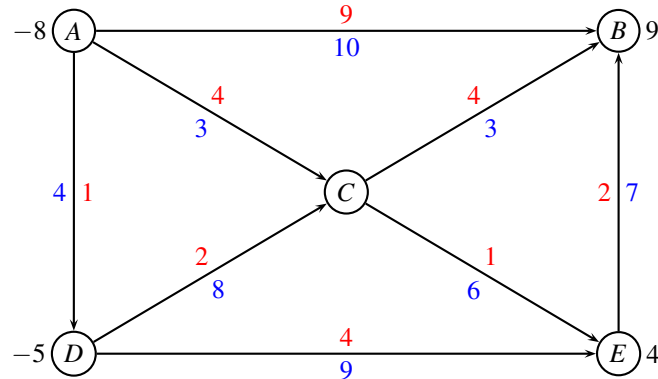
■

Problem 6: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Answer 6 A cut is $S = \{s, 2, 3, 4\}$ and $T = \{5, 6, t\}$. The capacity of the cut is 28. The flow in the solution is also 28. ■

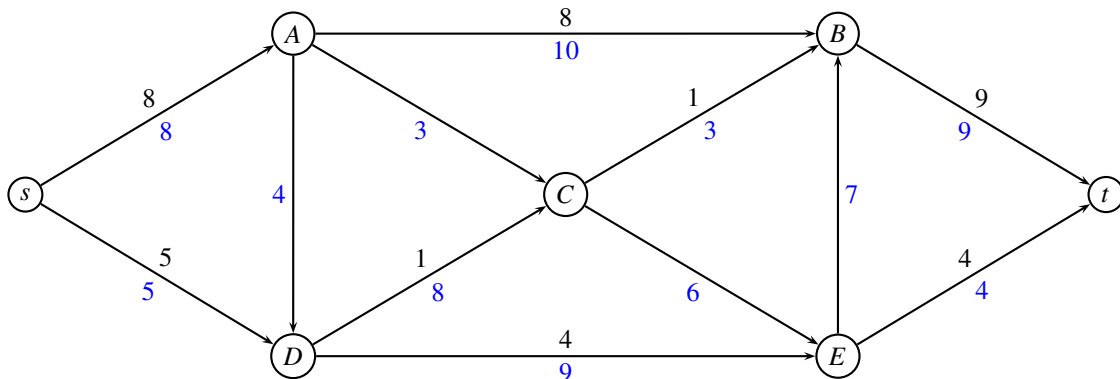
Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Problem 7: (small) Find first a legal start solution by solving an appropriate max-flow problem. ■

Answer 7 We add two extra nodes s, t , and solve a maximum-flow problem as follows: (blue values are capacity, black values are flow)

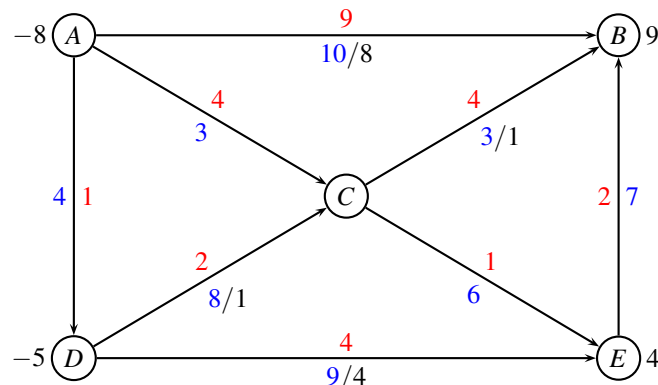


■

It is sufficient to draw the max-flow problem and to show the solution. No details about how the max-flow problem is solved are needed.

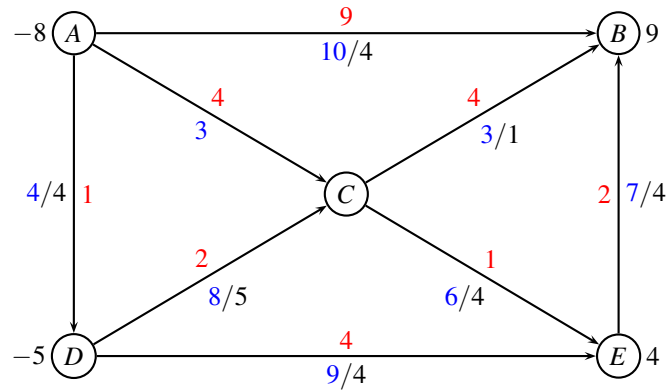
Problem 8: (large) Use the augmenting cycle algorithm on the above network. In each iteration, write down which augmenting cycle was used. ■

Answer 8 We start with:



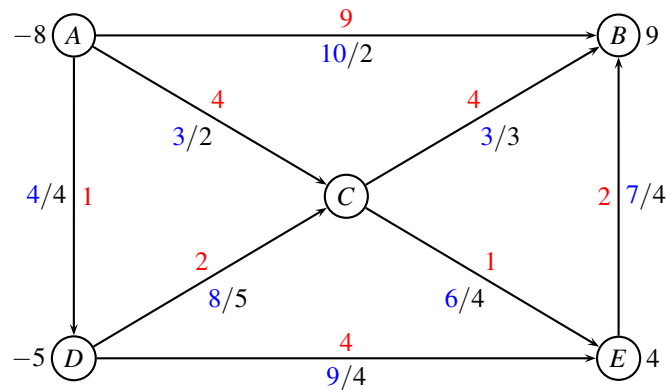
The cost of the initial solution is $72 + 4 + 2 + 16 = 94$.

An augmenting cycle is $ADCEBA$ with cost $1 + 2 + 1 + 2 - 9 = -3$. Maximum capacity is $\min\{4, 7, 6, 7, 8\} = 4$. After sending the 4 units the new graph becomes:



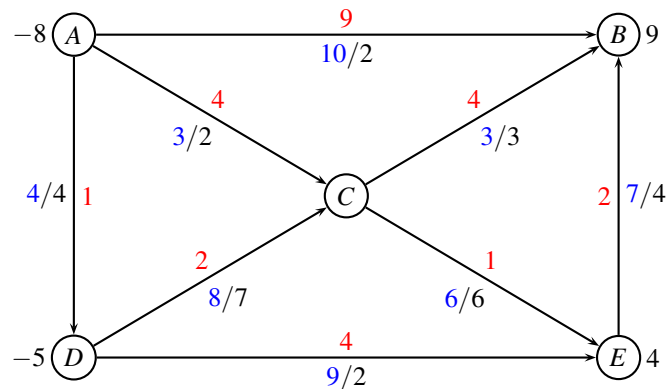
The cost of the solution is $36 + 4 + 4 + 8 + 10 + 4 + 16 = 82$.

An augmenting cycle is ACB with cost $4 + 4 - 9 = -1$. Maximum capacity is $\min\{3, 2, 8\} = 2$. Sending 2 units along the cycle:



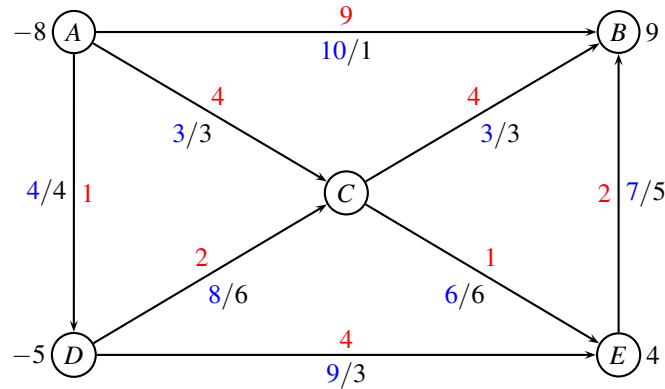
The cost of the solution is $18 + 8 + 12 + 4 + 8 + 10 + 4 + 16 = 80$.

An augmenting cycle is DCE with cost $2 + 1 - 4 = -1$. Maximum capacity is $\min\{3, 2, 4\} = 2$. Sending 2 units along the cycle:



The cost of the solution is $18 + 8 + 12 + 4 + 8 + 14 + 6 + 8 = 78$.

An augmenting cycle is $ACDEBA$ with cost $4 - 2 + 4 + 2 - 9 = -1$. Maximum capacity is $\min\{1, 5, 5, 3, 2\} = 1$. Sending 1 units along the cycle:



The cost of the solution is $9 + 12 + 12 + 4 + 10 + 12 + 6 + 12 = 77$. No more augmenting cycles can be found, so the solution is optimal.

Checking with CPLEX

```

minimize
  9 xAB + 4 xAC + 1 xAD + 4 xCB + 1 xCE + 2 xDC + 4 xDE + 2 xEB

subject to
  A: - xAD - xAB - xAC = -8
  B: xAB + xCB + xEB = 9
  C: xAC + xDC - xCB - xCE = 0
  D: xAD - xDC - xDE = -5
  E: xCE + xDE - xEB = 4

  xAB <= 10
  xAC <= 3
  xAD <= 4
  xCB <= 3
  xCE <= 6
  xDC <= 8
  xDE <= 9
  xEB <= 7

```

end

Dual simplex - Optimal: Objective = 7.7000000000e+01
 Solution time = 0.00 sec. Iterations = 3 (0)

Variable Name	Solution Value
xAB	1.000000
xAC	3.000000
xAD	4.000000
xCB	3.000000
xCE	6.000000
xDC	6.000000
xDE	3.000000
xEB	5.000000

■

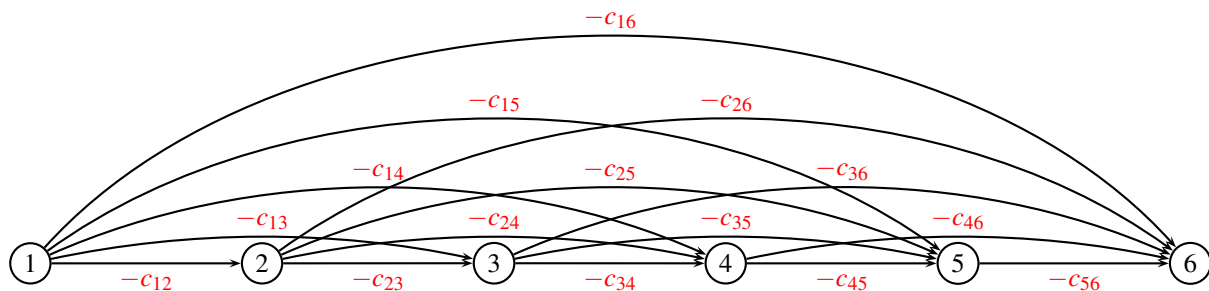
No details about how you found an augmenting cycle are needed.

Text formatting

The well-known document processing program LATEX uses an optimization procedure to decompose a paragraph into several lines so that when lines are left- and right-adjusted, the appearance of the paragraph will be the most attractive. Suppose that a paragraph consists of n words and that each word is assigned a sequence number. Let $c_{ij} > 0$ denote the attractiveness of a line if it begins with the word i and ends with word $j - 1$. Bigger values of c_{ij} are better. The program LATEX uses formulas to calculate the value of each c_{ij} , and we can assume they are given.

Problem 9: (large) Given the c_{ij} values, show how to formulate the problem of decomposing the paragraph into several lines of text in order to maximize the total attractiveness (of all lines) as a shortest path problem. ■

Answer 9 We get the graph:



Problem 10: (small) Does the graph have any negative cycles? ■

Answer 10 The figure is a DAG (directed acyclic graph). This means that all edges are pointing forward, hence no cycles are possible. ■

Problem 11: (medium) Propose an algorithm for solving the above problem. Try to find an algorithm with the best possible running time (in big-Oh notation). ■

Answer 11 Due to the negative edge weights, we cannot use Dijkstra which has the attractive running time $O(m \log n)$, where m is the number of edges and n is the number of nodes.

However, since we have a DAG, we can use dynamic programming, considering the nodes from left to right. In each step we find:

$$y_j = \min_{i:(i,j) \in E} \{y_i + w_{ij}\}$$

Since all y_i values are calculated before we consider y_j , the recursion is valid. The running time is $O(n + m)$, where n is the number of nodes and m is the number of edges. ■

Hotel planning

A hotel has received a number of requests for the following week as shown in the following table

start\end	Wed	Thu	Fri
Mon	2	3	-
Tue	4	3	5
Wed	-	4	3

Each entry in the table gives the number of guests requesting a room from the "start" day to the "end" day. A dash indicates that no requests have been given. A stay may not be split.

The hotel gets 2\$ for a 1-day stay, 5\$ for a 2-day stay, and 8\$ for a 3-day stay. The hotel wants to maximize the revenue. However, only the following number of rooms is available each of the nights:

Mon-Tue	Tue-Wed	Wed-Thu	Thu-Fri
5	6	8	7

Problem 12: (xlarge) Formulate the problem as a minimum cost flow problem by drawing a graph $G = (V, E)$ where each edge (i, j) has a cost c_{ij} , and an upper bound on the flow u_{ij} . Moreover, every node i has a demand b_i which is negative for a source, and positive for a sink. ■

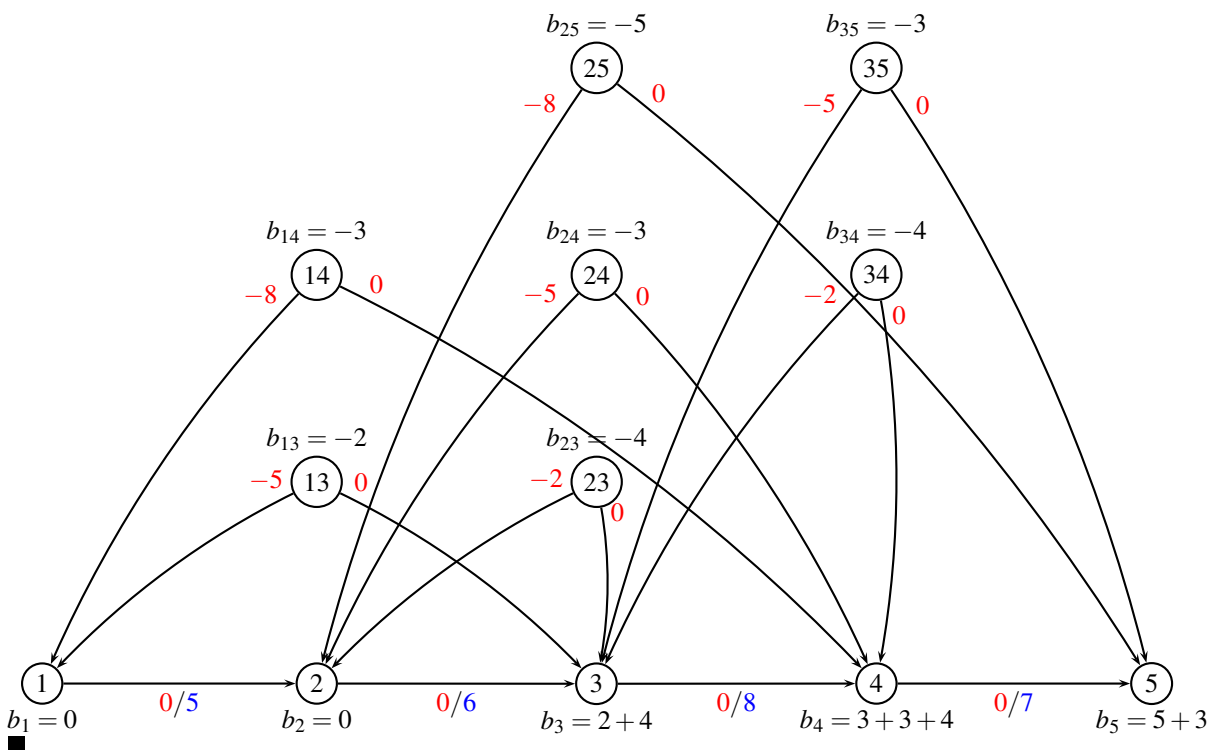
Answer 12 Let nodes 1, 2, 3, 4, 5 correspond to the week days considered. An edge from day i to day $i + 1$ has the capacity corresponding to the number of available rooms.

For every rental period from day i to day j we create a node ij . The node ij is connected to node i with an edge having cost $-p$ where p is the revenue of a stay from day i to day j . The edge has infinite capacity. Moreover, the node ij is connected to node j with an edge having cost 0 and infinite capacity.

Every node ij has a demand b_{ij} corresponding to the minus number of requests for a stay from day i to day j .

Every node i corresponding to a week day, has a demand b_i corresponding to the sum of all requests ending day i . E.g. Wednesday has a demand $b_3 = 2 + 4$, and Thursday has a demand $b_4 = 3 + 3 + 4$.

This leads to the following graph (capacities are blue, costs are red. If no capacity is given, then the capacity is infinite).



Hint: See application 9.4 in Ahuja, Magnanti, Orlin "Network Flows". (Available on campusnet).

Problem 13: (medium) Solve the problem with CPLEX or any other solver. Enclose a copy of the MIP model, and report the optimal solution. ■

Answer 13 We use the minimum cost flow formulation from above:

```

minimize
- 8 x14_1 - 5 x13_1 - 8 x25_2 - 5 x24_2 - 2 x23_2 - 5 x35_3 - 2 x34_3

subject to

- x14_1 - x14_4 = -3
- x13_1 - x13_3 = -2
- x25_2 - x25_5 = -5
- x24_2 - x24_4 = -3
- x23_2 - x23_3 = -4
- x35_3 - x35_5 = -3
- x34_3 - x34_4 = -4

x13_1 + x14_1 - x1_2 = 0
x23_2 + x24_2 + x25_2 + x1_2 - x2_3 = 0
x13_3 + x23_3 + x35_3 + x34_3 + x2_3 - x3_4 = 6
x14_4 + x24_4 + x34_4 + x3_4 - x4_5 = 10
x25_5 + x35_5 + x4_5 = 8

x1_2 <= 5
x2_3 <= 6
x3_4 <= 8
x4_5 <= 7

end

```

```

CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 4 rows and 0 columns.
Aggregator did 8 substitutions.
Reduced LP has 4 rows, 10 columns, and 20 nonzeros.
Presolve time = 0.00 sec.
Initializing dual steep norms . . .

```

```

Iteration log . . .
Iteration: 1 Dual objective = -71.000000

```

```

Dual simplex - Optimal: Objective = -6.0000000000e+01
Solution time = 0.00 sec. Iterations = 3 (0)

```

```

CPLEX> dis sol var -
Variable Name      Solution Value
x14_1              1.000000
x13_1              1.000000
x25_2              4.000000
x35_3              3.000000
x14_4              2.000000
x13_3              1.000000
x25_5              1.000000
x24_4              3.000000
x23_3              4.000000
x34_4              4.000000
x1_2               2.000000
x2_3               6.000000
x3_4               8.000000
x4_5               7.000000
All other variables in the range 1-18 are 0.

```

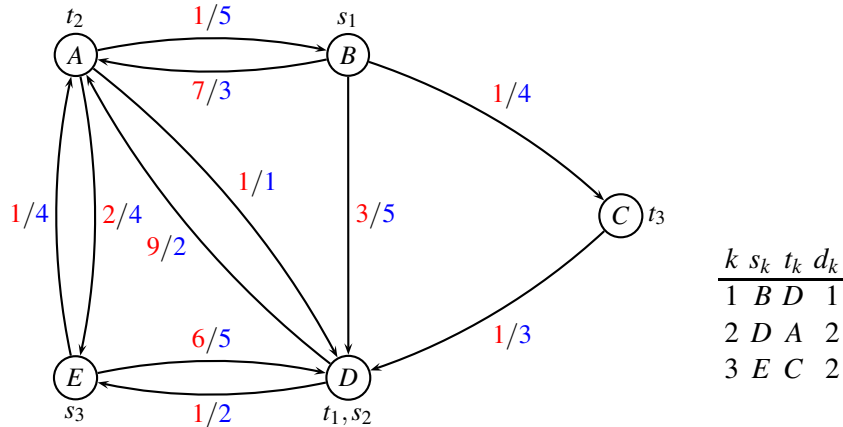
So, the solution is

start\end	Wed	Thu	Fri
Mon	1	1	-
Tue	0	0	4
Wed	-	0	3

With a total profit of 60\$. ■

Multicommodity flow

Consider the following unsplittable multicommodity flow problem where each edge has a cost and a capacity c_{ij}/u_{ij} . There are three commodities as indicated in the table, each commodity k having source s_k , terminal t_k and demand d_k .



A path formulation of the problem is given in the last page of the appendix.

Comment: In the following questions, the problem should be solved in the unsplittable version, but since it was not clear from the formulation, both splittable and unsplittable versions are accepted.

Problem 14: (small) Fill in the objective value in the path formulation, and solve the problem to LP-optimality and IP-optimality. Notice that the number of commodities flown is handled in the last constraints, as opposed to the project assignment. ■

Answer 14 (Unsplittable) We have the paths

number	commodity k	route r	d_k	$\sum c_{ij}$	cost
1	1	B→A→D	1	8	8
2	1	B→A→E→D	1	15	15
3	1	B→C→D	1	2	2
4	1	B→D	1	3	3
5	2	D→A	2	9	18
6	2	D→E→A	2	2	4
7	3	E→A→B→C	2	3	6
8	3	E→D→A→B→C	2	17	34

We modify the model from a splittable to an unsplittable problem:

```

minimize
+8x1+15x2+2x3+3x4+18x5+4x6+6x7+34x8
subject to
AB:                                +2x7+2x8 <= 5
AD: +1x1                            <= 1
AE:   +1x2                          <= 4
BA: +1x1+1x2                        <= 3
BC:   +1x3                          +2x7+2x8 <= 4
BD:   +1x4                          <= 5
CD:   +1x3                          <= 3
DA:   +2x5                          +2x8 <= 2
DE:   +2x6                          <= 2
EA:   +2x6+2x7                      <= 4
ED:   +1x2                          +2x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 1
K3: +x7+x8 >= 1

```

```

integer
x1 x2 x3 x4 x5 x6 x7 x8
end

```

The optimal solution is:

```

CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 5 rows and 0 columns.
Aggregator did 1 substitutions.
Reduced LP has 8 rows, 7 columns, and 19 nonzeros.
Presolve time = 0.00 sec.

Iteration log . . .
Iteration: 1 Dual objective = 6.000000

Dual simplex - Optimal: Objective = 1.2000000000e+01
Solution time = 0.00 sec. Iterations = 2 (0)

```

```

CPLEX> dis sol var -
Variable Name      Solution Value
x3                 1.000000
x6                 1.000000
x7                 1.000000
All other variables in the range 1-8 are 0.

```

■

Answer 14 (Splittable)

```

minimize
+8x1+15x2+2x3+3x4+9x5+2x6+3x7+17x8
subject to
AB:                                +x7+x8 <= 5
AD: +x1                            <= 1
AE:   +x2                          <= 4
BA: +x1+x2                        <= 3
BC:   +x3                          +x7+x8 <= 4
BD:   +x4                          <= 5
CD:   +x3                          <= 3
DA:   +x5                          +x8 <= 2
DE:   +x6                          <= 2
EA:   +x6+x7                      <= 4
ED:   +x2                          +x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 2

```

```

K3: +x7+x8 >= 2

end

CPLEX> read i
Problem 'i.lp' read.
Read time = 0.03 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 5 rows and 0 columns.
Aggregator did 1 substitutions.
Reduced LP has 8 rows, 7 columns, and 19 nonzeros.
Presolve time = 0.00 sec.

Iteration log . . .
Iteration: 1 Dual objective = 10.000000

Dual simplex - Optimal: Objective = 1.2000000000e+01
Solution time = 0.02 sec. Iterations = 2 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
x3                  1.000000
x6                  2.000000
x7                  2.000000
All other variables in the range 1-8 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
DE                   -7.000000
K1                   2.000000
K2                   9.000000
K3                   3.000000
All other dual prices in the range 1-14 are 0.

```

■

Problem 15: (large) Run column generation starting from columns corresponding to the variables x_1, x_5, x_7 . ■

Answer 15 (Unsplittable) When we use column generation we do not know all columns, and hence we should start with columns x_1, x_5, x_7 , and gradually find the next columns.

However, the present problem is very small, and we know all columns in advance, so we can add constraints $x_i \leq 0$ to all columns not being used.

First iteration

```

minimize
+8x1+15x2+2x3+3x4+18x5+4x6+6x7+34x8
subject to
AB:                +2x7+2x8 <= 5
AD: +1x1           <= 1
AE:   +1x2         <= 4
BA: +1x1+1x2       <= 3
BC:   +1x3         +2x7+2x8 <= 4
BD:           +1x4 <= 5
CD:   +1x3         <= 3
DA:           +2x5   +2x8 <= 2
DE:           +2x6   <= 2
EA:           +2x6+2x7 <= 4
ED:   +1x2         +2x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 1
K3: +x7+x8 >= 1

```

```

\\ P1: x1 <= 0
P2: x2 <= 0
P3: x3 <= 0
P4: x4 <= 0
\\ P5: x5 <= 0
P6: x6 <= 0
\\ P7: x7 <= 0
P8: x8 <= 0
end

CPLEX> read m1
Problem 'm1.lp' read.
Read time = 0.00 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 19 rows and 8 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec. (0.01 ticks)

Dual simplex - Optimal: Objective = 3.2000000000e+01
Solution time = 0.01 sec. Iterations = 0 (0)
Deterministic time = 0.01 ticks (0.81 ticks/sec)

```

```

CPLEX> dis sol var -
Variable Name      Solution Value
x1                  1.000000
x5                  1.000000
x7                  1.000000
All other variables in the range 1-8 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
K1                   8.000000
K2                  18.000000
K3                   6.000000
P3                  -6.000000
P4                  -5.000000
P6                  -14.000000
All other dual prices in the range 1-19 are 0.

```

Second iteration

```

minimize
+8x1+15x2+2x3+3x4+18x5+4x6+6x7+34x8
subject to
AB:                +2x7+2x8 <= 5
AD: +1x1           <= 1
AE:   +1x2         <= 4
BA: +1x1+1x2       <= 3
BC:   +1x3         +2x7+2x8 <= 4
BD:           +1x4 <= 5
CD:   +1x3         <= 3
DA:           +2x5   +2x8 <= 2
DE:           +2x6   <= 2
EA:           +2x6+2x7 <= 4
ED:   +1x2         +2x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 1
K3: +x7+x8 >= 1

```

```

\\ P1: x1 <= 0
P2: x2 <= 0
P3: x3 <= 0
P4: x4 <= 0
\\ P5: x5 <= 0
\\ P6: x6 <= 0

```



```

\\ P7: x7 <= 0
P8: x8 <= 0
end

CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 18 rows and 8 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec. (0.01 ticks)

Dual simplex - Optimal: Objective = 1.8000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)
Deterministic time = 0.01 ticks (10.90 ticks/sec)

```

```

CPLEX> dis sol var -
Variable Name      Solution Value
x1                  1.000000
x6                  1.000000
x7                  1.000000
All other variables in the range 1-8 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
DE                   -7.000000
K1                   8.000000
K2                  18.000000
K3                   6.000000
P3                   -6.000000
P4                   -5.000000
All other dual prices in the range 1-18 are 0.

```

Third iteration

```

minimize
+8x1+15x2+2x3+3x4+18x5+4x6+6x7+34x8
subject to
AB:                +2x7+2x8 <= 5
AD: +1x1           <= 1
AE:   +1x2         <= 4
BA: +1x1+1x2       <= 3
BC:   +1x3         +2x7+2x8 <= 4
BD:   +1x4         <= 5
CD:   +1x3         <= 3
DA:   +2x5         +2x8 <= 2
DE:   +2x6         <= 2
EA:   +2x6+2x7     <= 4
ED:   +1x2         +2x8 <= 5

```

```

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 1
K3: +x7+x8 >= 1

```

```

\\ P1: x1 <= 0
P2: x2 <= 0
\\ P3: x3 <= 0
P4: x4 <= 0
\\ P5: x5 <= 0
\\ P6: x6 <= 0
\\ P7: x7 <= 0
P8: x8 <= 0
end

```

```

CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 17 rows and 8 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec. (0.01 ticks)

```

Dual simplex - Optimal: Objective = 1.2000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)
Deterministic time = 0.01 ticks (10.57 ticks/sec)

```
CPLEX> dis sol var -
Variable Name      Solution Value
x3                  1.000000
x6                  1.000000
x7                  1.000000
All other variables in the range 1-8 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
DE                   -7.000000
K1                   2.000000
K2                   18.000000
K3                   6.000000
All other dual prices in the range 1-17 are 0.
```

■

Answer 15 (Unsplittable, not all columns known) If we did not know all columns in advance, we had to generate them gradually as follows:

First iteration

```
minimize
    +8x1      +18x5  +6x7
subject to
AB:          +2x7    <= 5
AD: +1x1          <= 1
AE:          <= 4
BA: +1x1          <= 3
BC:          +2x7    <= 4
BD:          <= 5
CD:          <= 3
DA:          +2x5    <= 2
DE:          <= 2
EA:          +2x7    <= 4
ED:          <= 5

K1: +x1 >= 1
K2: +x5 >= 1
K3: +x7 >= 1
end
```

```
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 14 rows and 3 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec. (0.00 ticks)
```

Dual simplex - Optimal: Objective = 3.2000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)
Deterministic time = 0.01 ticks (6.24 ticks/sec)

```
CPLEX> dis sol var -
Variable Name      Solution Value
x1                  1.000000
x5                  1.000000
x7                  1.000000
CPLEX> dis sol dua -
Constraint Name      Dual Price
K1                   8.000000
K2                   18.000000
K3                   6.000000
All other dual prices in the range 1-14 are 0.
```

Second iteration

```
minimize
    +8x1      +18x5+4x6+6x7
subject to
AB:          +2x7      <= 5
AD: +1x1      <= 1
AE:          <= 4
BA: +1x1      <= 3
BC:          +2x7      <= 4
BD:          <= 5
CD:          <= 3
DA:          +2x5      <= 2
DE:          +2x6      <= 2
EA:          +2x6+2x7   <= 4
ED:          <= 5

K1: +x1 >= 1
K2: +x5+x6 >= 1
K3: +x7 >= 1
end
```

```
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 14 rows and 4 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec. (0.00 ticks)

Dual simplex - Optimal: Objective = 1.8000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)
Deterministic time = 0.01 ticks (6.76 ticks/sec)
```

```
CPLEX> dis sol var -
Variable Name      Solution Value
x1                  1.000000
x6                  1.000000
x7                  1.000000
All other variables in the range 1-4 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price
DE                  -7.000000
K1                   8.000000
K2                  18.000000
K3                   6.000000
All other dual prices in the range 1-14 are 0.
```

Third iteration

```
minimize
    +8x1    +2x3    +18x5+4x6+6x7
subject to
AB:          +2x7      <= 5
AD: +1x1      <= 1
AE:          <= 4
BA: +1x1      <= 3
BC:          +1x3      +2x7   <= 4
BD:          <= 5
CD:          +1x3      <= 3
DA:          +2x5      <= 2
DE:          +2x6      <= 2
EA:          +2x6+2x7   <= 4
ED:          <= 5

K1: +x1+x3 >= 1
K2: +x5+x6 >= 1
K3: +x7 >= 1
end
```

```

CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 14 rows and 5 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec. (0.00 ticks)

Dual simplex - Optimal: Objective = 1.2000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)
Deterministic time = 0.01 ticks (7.47 ticks/sec)

```

```

CPLEX> dis sol var -
Variable Name          Solution Value
x3                      1.000000
x6                      1.000000
x7                      1.000000
All other variables in the range 1-5 are 0.
CPLEX> dis sol dua -
Constraint Name         Dual Price
DE                      -7.000000
K1                      2.000000
K2                      18.000000
K3                      6.000000
All other dual prices in the range 1-14 are 0.

```

■

Answer 15 (Splittable)

First iteration

```

minimize
+8x1+15x2+2x3+3x4+9x5+2x6+3x7+17x8
subject to
AB:                +x7+x8 <= 5
AD: +x1            <= 1
AE:  +x2          <= 4
BA: +x1+x2        <= 3
BC:  +x3          +x7+x8 <= 4
BD:      +x4      <= 5
CD:      +x3      <= 3
DA:      +x5      +x8 <= 2
DE:      +x6      <= 2
EA:      +x6+x7   <= 4
ED:  +x2          +x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 2
K3: +x7+x8 >= 2

\\ X1: x1 <= 0
X2: x2 <= 0
X3: x3 <= 0
X4: x4 <= 0
\\ X5: x5 <= 0
X6: x6 <= 0
\\ X7: x7 <= 0
X8: x8 <= 0

end

```

```

CPLEX> read i0.lp
Problem 'i0.lp' read.
Read time = 0.02 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 19 rows and 8 columns.

```

All rows and columns eliminated.

Presolve time = 0.00 sec.

Dual simplex - Optimal: Objective = 3.2000000000e+01

Solution time = 0.00 sec. Iterations = 0 (0)

CPLEX> dis sol var -

Variable Name	Solution Value
x1	1.000000
x5	2.000000
x7	2.000000

All other variables in the range 1-8 are 0.

CPLEX> dis sol dua -

Constraint Name	Dual Price
K1	8.000000
K2	9.000000
K3	3.000000
X3	-6.000000
X4	-5.000000
X6	-7.000000

All other dual prices in the range 1-19 are 0.

Second iteration

minimize

+8x1+15x2+2x3+3x4+9x5+2x6+3x7+17x8

subject to

AB: +x7+x8 <= 5

AD: +x1 <= 1

AE: +x2 <= 4

BA: +x1+x2 <= 3

BC: +x3 +x7+x8 <= 4

BD: +x4 <= 5

CD: +x3 <= 3

DA: +x5 +x8 <= 2

DE: +x6 <= 2

EA: +x6+x7 <= 4

ED: +x2 +x8 <= 5

K1: +x1+x2+x3+x4 >= 1

K2: +x5+x6 >= 2

K3: +x7+x8 >= 2

\\ X1: x1 <= 0

X2: x2 <= 0

X3: x3 <= 0

X4: x4 <= 0

\\ X5: x5 <= 0

\\ X6: x6 <= 0

\\ X7: x7 <= 0

X8: x8 <= 0

end

CPLEX> read i1.lp

Problem 'i1.lp' read.

Read time = 0.00 sec.

CPLEX> opt

Tried aggregator 1 time.

LP Presolve eliminated 18 rows and 8 columns.

All rows and columns eliminated.

Presolve time = 0.00 sec.

Dual simplex - Optimal: Objective = 1.8000000000e+01

Solution time = 0.00 sec. Iterations = 0 (0)

CPLEX> dis sol var -

Variable Name	Solution Value
x1	1.000000
x6	2.000000
x7	2.000000

All other variables in the range 1-8 are 0.

CPLEX> dis sol dua -

Constraint Name	Dual Price
DE	-7.000000
K1	8.000000
K2	9.000000
K3	3.000000
X3	-6.000000
X4	-5.000000

All other dual prices in the range 1-18 are 0.

Third iteration

```

minimize
+8x1+15x2+2x3+3x4+9x5+2x6+3x7+17x8
subject to
AB:          +x7+x8 <= 5
AD: +x1          <= 1
AE:  +x2          <= 4
BA: +x1+x2        <= 3
BC:  +x3          +x7+x8 <= 4
BD:  +x4          <= 5
CD:  +x3          <= 3
DA:  +x5          +x8 <= 2
DE:  +x6          <= 2
EA:  +x6+x7       <= 4
ED:  +x2          +x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 2
K3: +x7+x8 >= 2

\\ X1: x1 <= 0
X2: x2 <= 0
\\ X3: x3 <= 0
X4: x4 <= 0
\\ X5: x5 <= 0
\\ X6: x6 <= 0
\\ X7: x7 <= 0
X8: x8 <= 0

end

CPLEX> read i2.lp
Problem 'i2.lp' read.
Read time = 0.01 sec.
CPLEX> opt
Tried aggregator 1 time.
LP Presolve eliminated 17 rows and 8 columns.
All rows and columns eliminated.
Presolve time = 0.00 sec.

Dual simplex - Optimal: Objective = 1.2000000000e+01
Solution time = 0.00 sec. Iterations = 0 (0)

CPLEX> dis sol var -
Variable Name      Solution Value
x3                 1.000000
x6                 2.000000
x7                 2.000000
All other variables in the range 1-8 are 0.
CPLEX> dis sol dua -
Constraint Name      Dual Price

```

DE	-7.000000
K1	2.000000
K2	9.000000
K3	3.000000

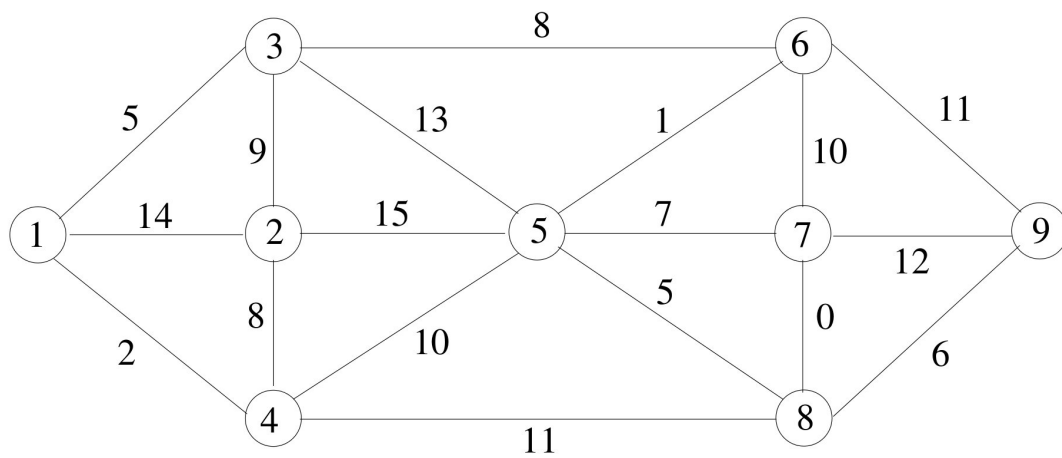
All other dual prices in the range 1-17 are 0.

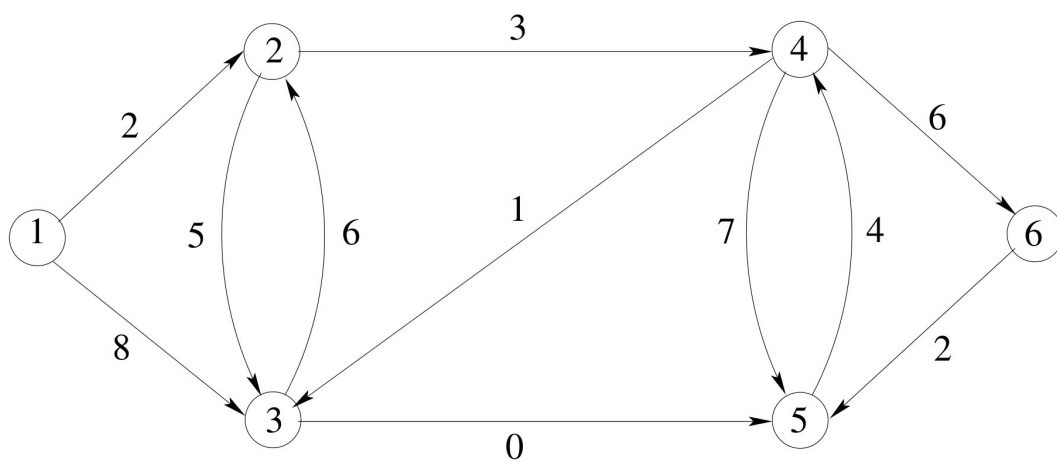
■

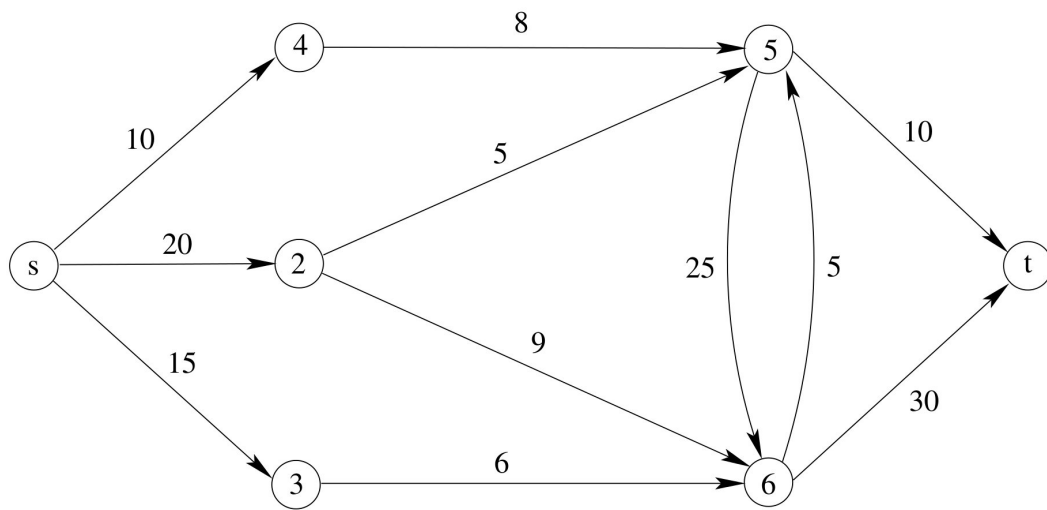
Hint: In the last question you can add a constraint $x_i \leq 0$ to all columns not being used. It is sufficient to show the input (written as an LP-problem) and the primal/dual solution in each iteration. You may use CPLEX or any other LP-solve. When reporting the input it should be written as a normal LP-problem.

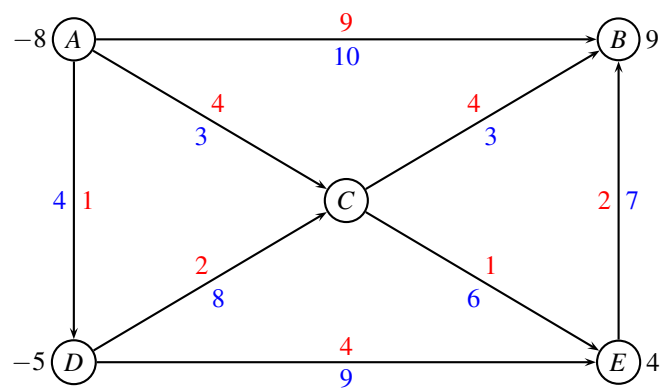
Appendix

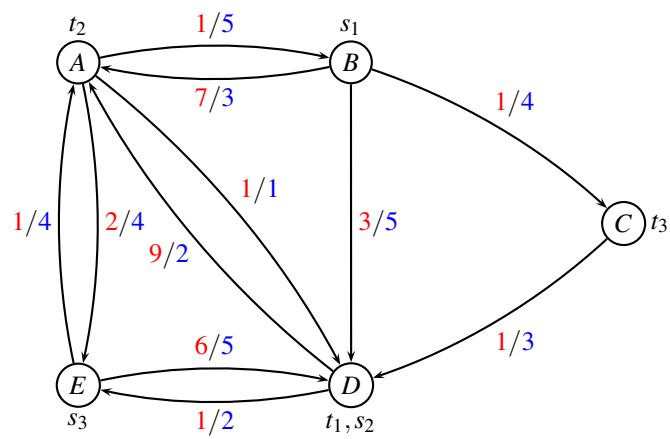
The figures on the following pages can be used when solving the problem. It is allowed to write by hand on the figures in your answer.











```

minimize
  + x1+ x2+ x3+ x4+ x5+ x6+ x7+ x8
subject to
AB:                                     +x7+x8 <= 5
AD: +x1                                <= 1
AE:   +x2                              <= 4
BA: +x1+x2                             <= 3
BC:   +x3                               +x7+x8 <= 4
BD:   +x4                               <= 5
CD:   +x3                               <= 3
DA:   +x5                               +x8 <= 2
DE:   +x6                               <= 2
EA:   +x6+x7                           <= 4
ED:   +x2                               +x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 2
K3: +x7+x8 >= 2

end

```

Take-home exam: December 19, 2012.

Course name: Network Optimization

Course no: 42115

Name	study number

I hereby declare that I have worked alone on the project assignment and not discussed it with anybody else.

Date

Signature
